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Robust Booster Disinfection Scheduling Using Incomplete Mixing Water Quality Model (EPANET-IMX)

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ABSTRACT

In the realm of practical water distribution systems (WDS), uncertainties in hydraulic and water quality modeling affect the management of WDS making it a multifaceted challenge. Furthermore, the incorporation of water quality considerations into WDS design and management has become paramount, necessitating the use of precise water quality models. The conventional water quality models employed in drinking WDS have faced inaccuracy due to their underlying assumption of instantaneous and complete mixing at junctions. To address this limitation, recent models such as EPANET-IMX (Incomplete Mixing Extension), EPANET-BAM, and AZRED have emerged, incorporating empirical equations to model the nuances of incomplete mixing. These advancements offer improved accuracy for water quality analysis within WDS. However, the presence of uncertainty in disinfectant reaction rates also presents an obstacle to achieving optimal water quality management. Within such systems, determining the scheduling of booster disinfectant dosages is a challenge. In response, this study seeks to determine the optimal dosage schedule for booster disinfectants while accounting for fluctuations in bulk reaction rate coefficients and acknowledging incomplete mixing at junctions. To tackle this uncertain optimization problem, robust optimization principles are employed. The study applies these principles to a small-scale network as an illustrative example, showcasing robust optimal schedules. Two water quality models, namely EPANET and EPANET-IMX, are utilized for water quality simulations. The resulting optimal schedules are compared and analyzed across all three models. It was observed that considering the incomplete mixing varied the optimal booster dosage by about 10%. The findings emphasized the importance of considering incomplete mixing in both water quality analysis and optimization endeavors.

INTRODUCTION

A water distribution system is comprehensively defined as a network comprising pipes, tanks, pumps, and valves that establish a connection between a water source and end-users, ensuring the delivery of high-quality water at sufficient pressures. Preliminary investigations pertaining to water distribution systems primarily focus on validating hydraulic parameters, encompassing considerations such as water demands and pressures. With an increasing awareness of the pivotal role played by water quality, water utilities are obligated to guarantee

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the provision of high-quality water to end-users. This commitment aligns with the sustainable development goals outlined by the United Nations (UN), with an ambitious target for achievement by the year 2030.

To ensure the optimal functioning of the Water Distribution System (WDS), water utilities necessitate to employ precise models of WDS in order to accurately estimate the state of water reaching the consumers. The majority of commercialized models or software, such as EPANET2.2, WaterGEMS (Version 10.04.00.108), or InfoWater Pro (2023.3), assumes complete and instantaneous mixing of water at cross junctions of drinking WDS to simplify the computational processes. This assumption, however, has been demonstrated to be inaccurate in various experimental and numerical studies (Shang et al., 2021, Shao et al., 2019, Romero-Gomez et al., 2008, Austin et al., 2008). It was observed that in some cases, inflows at cross junctions and double tee junctions not only do not mix completely but also bifurcate.

The degree of mixing is contingent upon multiple parameters associated with the incoming flows into the cross-junctions. Researchers developed water quality models to address the incomplete mixing. EPANET-IMX (Incomplete Mixing Extension) is the most recent empirical model considering incomplete mixing in cross junctions (Yousefian and Duchesne, 2024). EPANET-BAM (Bulk Advection Method) was developed based on considering only the advection term in mixing (Ho and Khalsa, 2008), , and AZRED (AZ stands for Arizona State and RED stands for network in Spanish) is an empirical model considering incomplete mixing. More details about these models and other existing incomplete mixing models were discussed in a research by Yousefian and Duchesne (2022). Also, the importance of considering incomplete mixing in different applications of water quality modelling was discussed in different studies (Mompremier et al., 2022, van Summeren et al., 2017, Romero-Gomez et al., 2011).

Another significant challenge faced by water utilities in the design and management of WDS is the uncertainties associated with system parameters. Pasha and Lansey (2005, 2010) examined the impact of uncertain sources like reaction coefficient and pipe diameter on the system's water quality analysis. They found that the bulk reaction coefficient has the most influence among all studied water quality parameters. The optimal booster chlorination problem faced by water utilities is notably affected by uncertainty in the bulk decay rate of chlorine. Köker et al. (2015) employed chance-constraint programming to address the optimal booster chlorination scheduling problem, while Wang and Zhu (2018, 2021) proposed an inexact two-stage chance-constrained programming approach. Boindala et al. (2023) introduced a robust optimization approach considering box uncertainty sets to bound the uncertain bulk decay coefficient for optimal booster chlorination schedules.

This study represents an initial exploration to evaluate the impact of considering the incomplete mixing phenomenon in optimizing the dosage of boosters chlorination. It contrasts optimal booster chlorination schedules derived from both complete mixing and incomplete mixing models for estimating residual chlorine levels at consumers, taking into account uncertainties associated with incomplete mixing. Utilizing a robust optimization approach introduced by Boindala et al. (2023), the study aims to obtain optimal schedules, incorporating assessments from both the complete mixing model (EPANET) and incomplete mixing model (EPANET-IMX) for residual chlorine estimates.

METHODOLOGY

The methodology for this study is divided into four stages namely initialization, simulation, optimization and analysis. A flow chart is provided in Figure 1.

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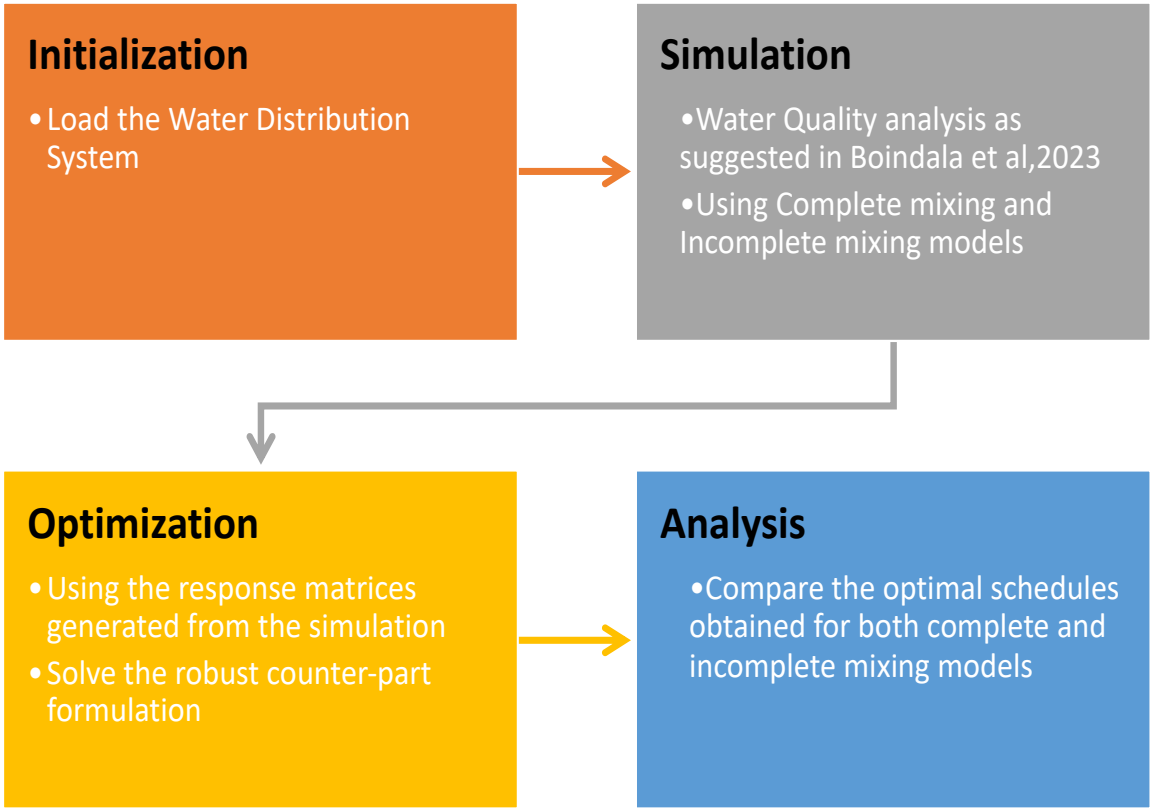

Figure 1. Methodology Flow Diagram

COMPLETE MIXING (EPANET)

The complete mixing model finds extensive application in water quality analysis software such as EPANET. This modelling approach assumes instantaneous and complete mixing within pipe junctions, thereby obviating the necessity for intricate numerical models. Its notable advantage lies in its ease of integration into water quality models, facilitating efficient and prompt computations.

Within the framework of this model, the concentration in the fluid exiting a junction is consistently uniform across all outlet pipes, contingent upon the flow-weighted concentrations entering the pipe. The model's foundation rests on the principle of solute mass balance, yielding a straightforward equation that delineates the concentration leaving the junction in any given outlet pipe.

$$C_{out} = \sum_{i=1}^{n} Q_i C_i \Big/ \sum Q_{out} = \sum_{i=1}^{n} Q_i C_i \Big/ \sum_{i=1}^{n} Q_i \tag{1}$$

where Q_i is the flow rate [L^3/T] entering the junction from pipe or source i; n is the number of inlet pipes or flow sources entering the junction; and C_i is the concentration [M/L^3] entering from each pipe or source to the junction. No storage is assumed to exist in the junction, so the total outlet flow rate ($\sum Q_{out}$)of the junction, is equal to the total inflow rate ($\sum_{i=1}^{n} Q_i$) into the junction.

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INCOMPLETE MIXING MODEL (EPANET-IMX)

EPANET-IMX considers incomplete mixing for cross junctions with two adjacent inflows and two adjacent outflows. For each hydraulic simulation time step, the flow rates in the legs of cross junctions are calculated. Then, based on the equations developed by Yousefian and Duchesne (2024), the concentrations in the outlets are estimated. EPANET-IMX was evaluated in the mini water distribution network laboratory of Institut National de la Recherche Scientifique (INRS) in Quebec City, Canada. This laboratory is equipped with two pumps (a 3 hp (Xylem-AquaBoost) and a 75 hp (Berkeley-B4EPBMS)), six pressure probes (Ashcroft-G2), twelve flow control valves with electric actuators (Assured Automation-P2R4 with S4 actuator), four pressure reducing valves (SINGER-106/206-PR) and nine electromagnetic flow meters (ModMAG-M2000) (Figure 2).

Figure 2. Mini water distribution network laboratory of INRS

All the settings of pumps and valves were set through a central computer (Honeywell-EBI R430 and Honeywell-Controller HC900) to increase the accuracy of applied conditions in experiments (Figure 3).

OPTIMAL BOOSTER CHLORINATION PROBLEM

Chlorine plays a pivotal role in maintaining satisfactory water quality within water distribution systems (WDS), acting as the primary disinfectant. To guarantee the absence of microorganisms, a minimum chlorine concentration requirement is imposed at consumer nodes. However, excessive chlorine levels can lead to the formation of disinfection by-products, posing

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health risks and imparting an unpleasant taste and odour to the water (Wang et al., 2018). Striking a balance between controlling pathogen growth and minimizing disinfection by-products is crucial for averting long-term health hazards (Boccelli et al., 1998).

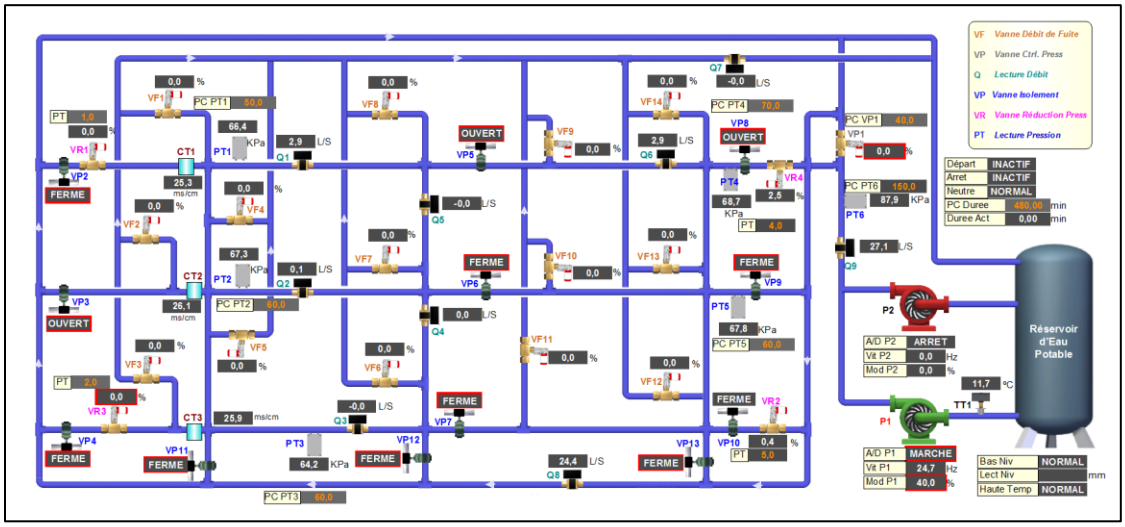

Figure 3. Control software of the laboratory in the central computer

The acceptable range of chlorine concentration varies globally, typically falling within 0.2 mg/l to 4.0 mg/l according to general guidelines (Goyal and Patel, 2017; Köker and Altan-Sakarya, 2015). To adhere to these limits, strategically positioned booster chlorination stations are deployed and optimally operated throughout the water distribution network.

In the context of water quality evaluation in WDS, residual chlorine concentration, estimated by models like EPANET, serves as a key metric. Incorporating the bulk reaction coefficient of chlorine into these models involves considering multiple factors such as contaminants, pipe material, water age, pH, temperature, and external influences. The accurate prediction of these parameters presents challenges, introducing uncertainty into the reaction coefficient. This uncertainty complicates the planning of an optimal disinfectant injection rate necessary for meeting water quality requirements over an extended operational period.

Addressing the optimal booster disinfection problem entails determining the precise amount of disinfectant required for injection, achieving an effective trade-off between controlling pathogen growth and minimizing disinfection by-products.

OPTIMIZATION

To obtain the optimal disinfection dosage, a robust optimization methodology proposed by Boindala et al. (2023) was used. The water quality models, both complete and incomplete mixing models were replaced with linear surrogate models using the principle of superposition proposed by Boccelli et al. (1998).

WATER QUALITY OBJECTIVE

For each water quality sensor location, the penalty associated with residual disinfectant concentration was evaluated based on penalty-based function (Kurek et al., 2012) as described in

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Figure 2. This is a different approach compared to conventional minimization of the mass injected into the system that does not restrict the water quality to be within the hard bounds of $[C_{Lower}, C_{Upper}]$ residual disinfection limits.

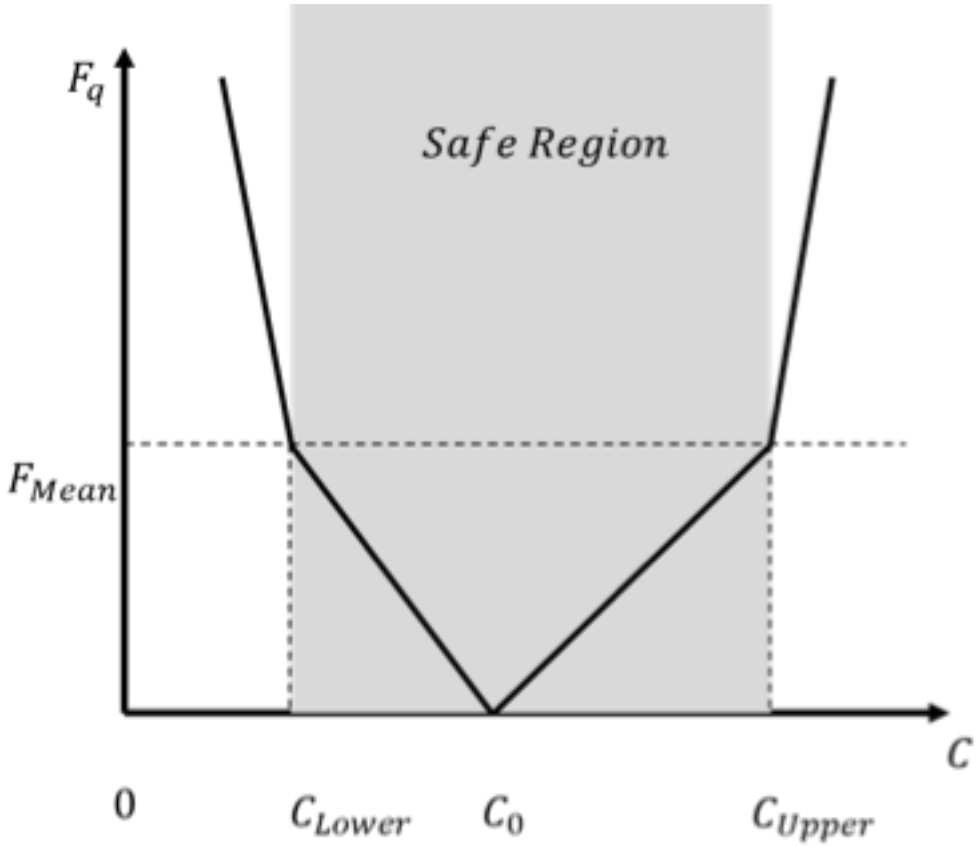

Figure 4. Penalty based objective function from Kurek et al. (2012)

The objective function used in this study for residual disinfectant concentration was as follows in Eq. (2):

$$\text{obj: } F(x) = \sum_{t=1}^{T} \sum_{ns=1}^{S} F_q(C_{ns}(t)) \tag{2}$$

where S is asset of sensor nodes where water quality is estimated, $F_q()$ is the water quality function explained in Figure-2, and $C_{ns}(t)$ is the residual concentration at sensor node ns for time instant $t \in T$. The sensor nodes and the water quality bounds can be altered according to the requirement. The residual disinfection function can be written as shown in Eq (3). Here f_i are the linear functions based on Figure 2 where, $c_1 = c_4 = F_{mean}, c_2 = c_3 = 0, d_1 = M1, d_2 = \frac{F_{mean}}{C_0 - C_{Lower}}, d_3 = -1 * \left(\frac{F_{mean}}{C_{Upper} - C_0}\right), d_4 = -1 * M2,$ and $C_1 = C_{Lower}, C_2 = C_3 = C_0, C_4 = C_{Upper}$. For this study, $F_{mean} = 10, M1 = 250, M2 = 100, C_0 = 0.5. C_{lower} = 0.2,$ and $C_{upper} = 4$ were used.

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$$F_q(C_{ns}(t)) = \max_{i=1,2,3,4} f_i \text{ where } f_i = c_i + d_i(C_i - C_{ns}) \tag{3}$$

By using Eq. (3), the booster disinfectant optimization problem can be written as follows in Eq. (4) and Eq. (5), where uncertian response matrix $\widetilde{B} \in U_B$. U_B is the response matrix uncertain set.

$$\min_x F(x) = \sum_{t=1}^{T} \sum_{ns=1}^{S} \max_{i=1,2,3,4} f_i(c_{ns}(t)) \tag{4}$$

such that:

$$c_{ns}(t) = \widetilde{B}_{ns}(t)x, \forall ns \in S, \forall \widetilde{B}_{ns} \in U_B; x \geq 0 \tag{5}$$

The maximum of the piece-wise linear function can be re-written as $\mathbb{P}$:

$$\min_x F(x) = \sum_{t=1}^{T} \sum_{ns=1}^{S} z_{t,ns}$$

such that:

$$\begin{gathered} f_1\big(c_{ns}(t)\big) \leq z_{t,ns} \\ f_2\big(c_{ns}(t)\big) \leq z_{t,ns} \\ f_3\big(c_{ns}(t)\big) \leq z_{t,ns} \\ f_4\big(c_{ns}(t)\big) \leq z_{t,ns} \\ c_{ns}(t) = \widetilde{B}_{ns}(t)x, \forall ns \in S, \forall \widetilde{B}_{ns} \in U_B \\ x \geq 0 \end{gathered} \tag{$\mathbb{P}$}$$

The robust reformulation of this can be expressed as ($\mathbb{RP}$):

$$\min_x F(x) = \sum_{t=1}^{T} \sum_{ns=1}^{S} z_{t,ns}$$

Such That:

$$c_i + d_i C_i + UB_{B_{ns,t}} \alpha_{i,t,ns} - LB_{B_{ns,t}} \beta_{i,t,ns} \leq z_{t,ns} \; \forall ns \in S, t \in T, i \in [1,2,3,4] \tag{$\mathbb{RP}$}$$

$$\beta_{i,t,ns} - d_i x - \alpha_{i,t,ns} = 0 \; \forall ns \in S, t \in T, i \in [1,2,3,4]$$

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$$x, \alpha_{i,t,ns}, \beta_{i,t,ns}, z_{t,ns} \geq 0 \; \forall ns \in S, t \in T, i \in [1,2,3,4]$$

The procedure for obtaining the robust counterpart is explained in Boindala et al. (2023).

CASE STUDY

In this research, the first example (NET-1) provided with the OWA-EPANET toolkit was selected as a case study. The hydraulic conditions (pipes, valves, pumps, demands, etc.) of this example (as provided in Open Water Analytics OWA-EPANET toolkit) were kept without any change. However, for the water quality conditions, the initial concentrations at all nodes and links was set to 0 and two junctions (J1 and J3), as shown in Fig 5, were selected as the mass booster sources. The quality tolerance was set to 0.01. Regarding the reaction terms, the first order bulk reaction and the first order wall reaction were employed. The global wall coefficient was set to -1.0 (mass/L/day) and the global bulk coefficient to -0.2(1/day) for one series of simulation and to -0.8(1/day) for the second series of simulation.

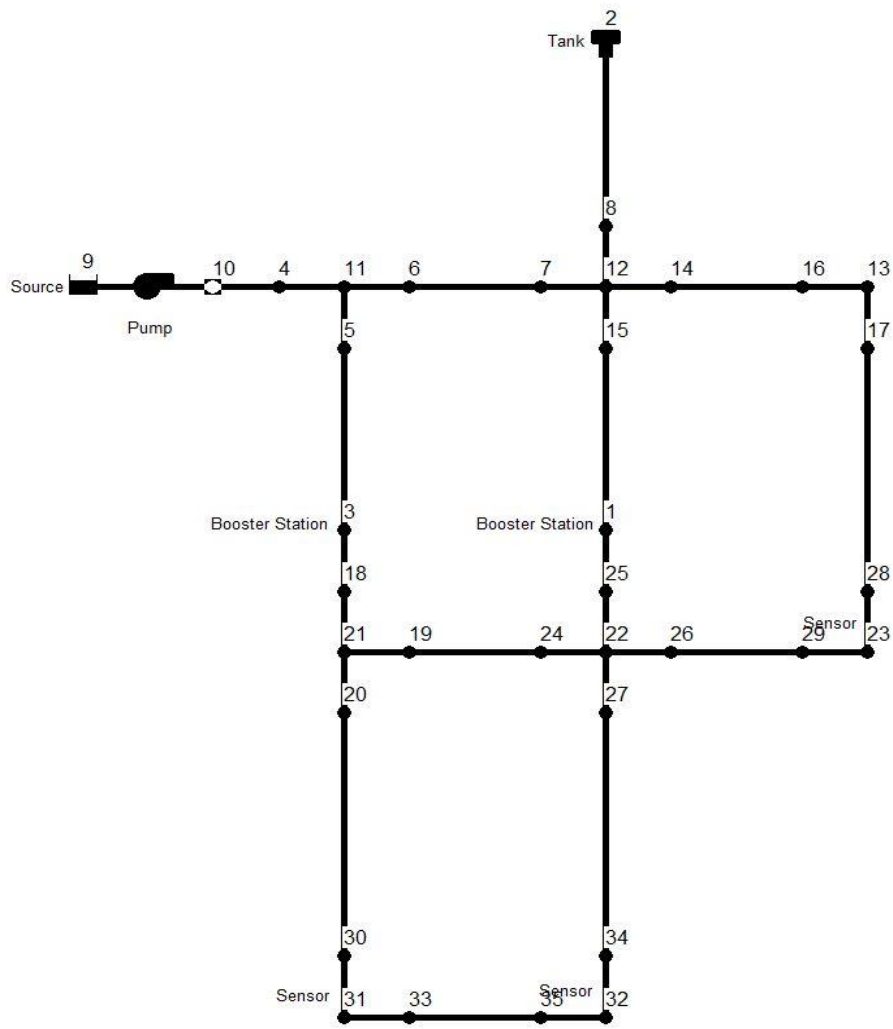

Figure 5. Schematic of case study

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Two sensors were supposed to be present in the network: at junctions 31 and 32. The concentrations of chlorine in these junctions were the main output of the simulations.

RESULTS

The assessment of water quality necessitates the consideration of defined ranges for chlorine concentration, with established lower and upper bounds set at 0.2 mg/l and 4 mg/l, respectively. A network simulation spanning 24 hours has been executed to compute water quality, facilitating the estimation of response matrices corresponding to the lower and upper bounds. The bulk reaction coefficient of chlorine is constrained within a box uncertainty set defined as [-0.8(1/day), -0.2 (1/day)].

The findings of the study are presented through graphs illustrating the optimal booster schedules for both Complete-mixing and incomplete mixing models, as depicted in Figures 6 and 7. Additionally, a comparative analysis of the total amount of optimal chlorine injected into the system for both models is illustrated in Figure 8. These visual representations offer insights into the impact of mixing models on optimal booster schedules and the overall chlorine injection strategy within the system.

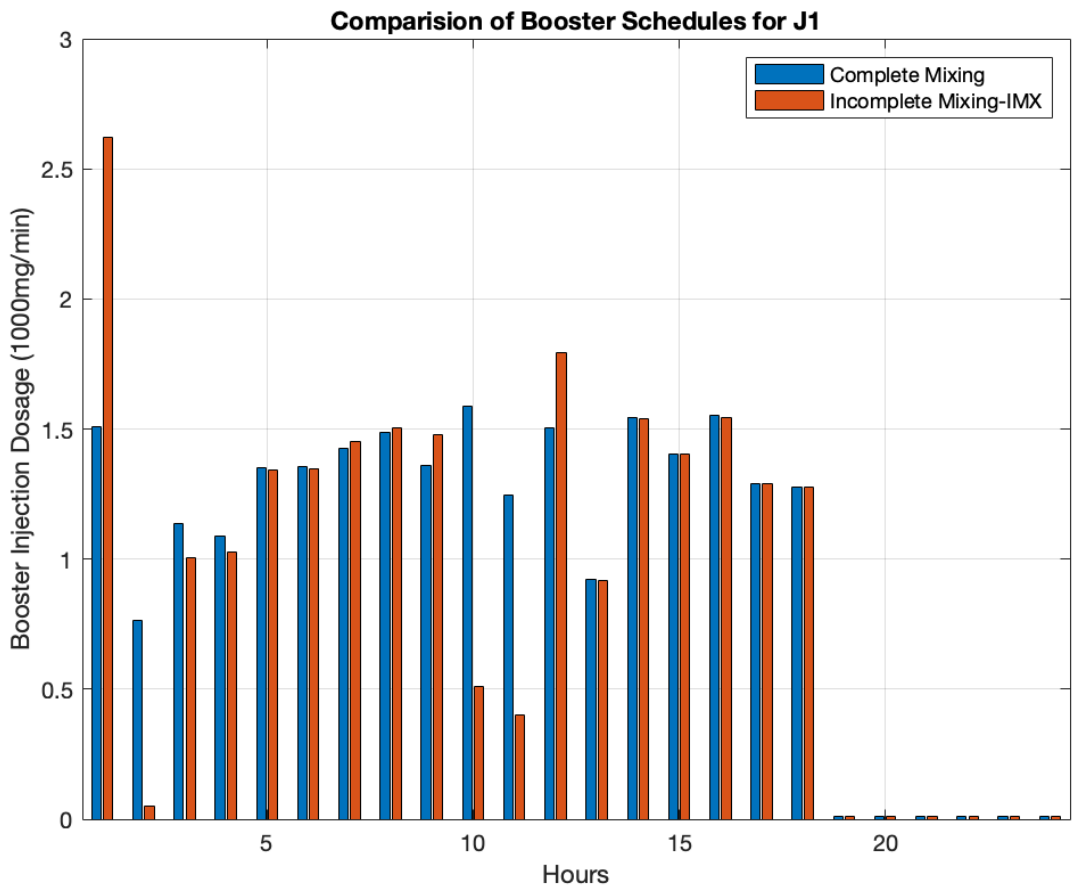

Figure 6. Optimal booster schedule for Booster J1

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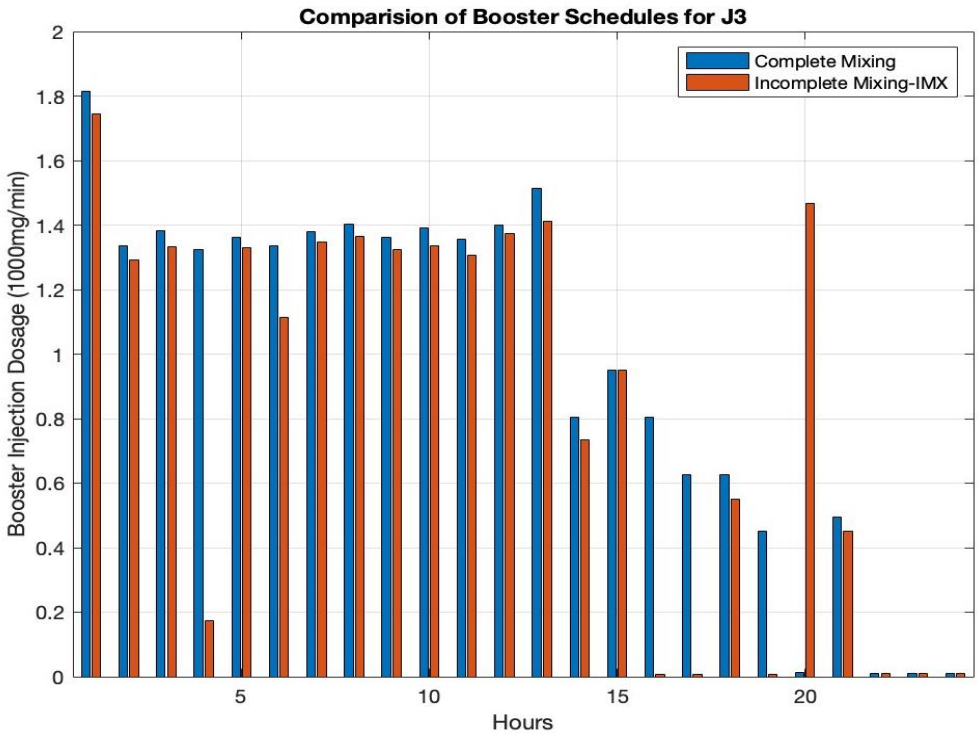

Figure 7. Optimal Booster Schedules for Booster Station J3

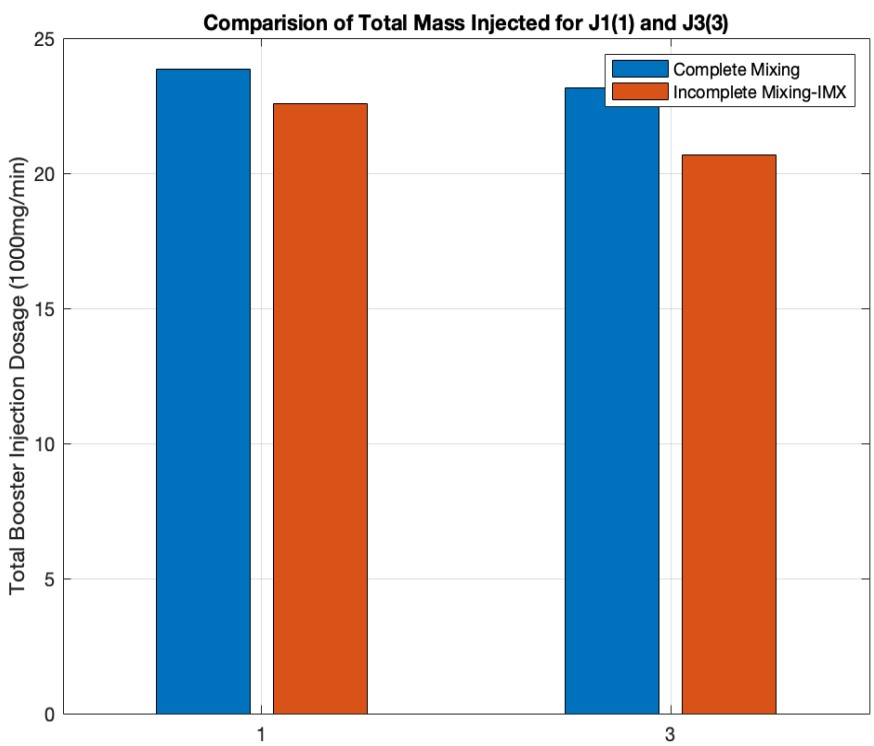

Figure 8. Comparison of total mass injected into the system

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The comparative graphs of optimal booster chlorination schedules (Figure 6 and 7) reveal an increase in the initial hours(1-4) at J1 and the last few hours(16,17,19,20) at J3 booster stations within a few hours. These results can be explained by the mixing in cross junction 22. Based on the demand scenario applied in the first example of EPANET 2.2, the cross junction 22 had two adjacent inflows and two adjacent outflows from the time 0 to 12 hour and from the time 22 to 24 hour. Accordingly, in EPANET-IMX, when there are two adjacent inlets and two adjacent outlets, the mixing would be incomplete. However, when there are two opposing inlets and two opposing outlets (from time 12 to 22 hour in the current case study), the mixing would be complete. Therefore, when the booster station at J1 injected the chlorine, in the first time steps, the results of mixing were different in EPANET 2.2 and EPANET-IMX and therefore, the optimum dosage was different for these time steps in complete and incomplete mixing simulations. Similarly, for the booster station at J3, these differences were observed in the last time steps as the distance between J3 and the cross junction 25 was long and it took time until the boosted chlorine reached to the cross junction 25 and the sensor at junction 23.

Although the injection dosages remain largely consistent between models for the majority of hours, discernible distinctions emerge in the optimal schedules during specific time intervals. This underscores the significance of accounting for incomplete mixing at junctions when addressing water-quality-based objectives in the optimization problem. Furthermore, the comprehensive view presented in Figure 8 indicates an overall reduction in the mass injected into the system with the incomplete mixing model in comparison to the Complete mixing model. This observation emphasizes the impact of considering incomplete mixing on the overall mass balance within the system.

CONCLUSION

This study, in its preliminary phase, aims to assess the importance of integrating incomplete mixing models into the optimization of water quality objectives. The research involves a comparative examination of optimal booster schedules derived from both a complete mixing model (EPANET) and an incomplete mixing model (EPANET-IMX) in the context of water quality analysis. Further investigation is imperative to elucidate the exact underlying reasons for the changes observed in optimal schedules.

To achieve a more comprehensive understanding, an extension of the comparative analysis is recommended, encompassing additional incomplete mixing models such as AZRED and EPANET-BAM. This expanded analysis would contribute to a more nuanced comprehension of their respective impacts on optimization outcomes.

Moreover, it is suggested that the study should broaden its scope to include other water-quality objectives, such as the minimization of water age. This extension is essential for gaining a deeper insight into the implications of considering incomplete mixing models at junctions in water quality analysis and, consequently, in the pursuit of water-quality objectives.

From this study we can clearly see that using conventional complete mixing for water quality analysis and optimal water distribution system design and operation could lead to sub-optimal solutions when applied practically.

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